

Combinations

List all possible combinations.

1) T, V, W, taken two at a time

2) A, B, C, D, taken two at a time

3) ☺, ☀, ♥, ▲, taken three at a time

4) 4, 5, 6, 7, taken four at a time

Evaluate each expression.

5) ${}_{22}C_{20}$

6) ${}_{11}C_8$

7) ${}_{12}C_8$

8) ${}_{25}C_{23}$

9) ${}_{24}C_5$

10) ${}_{17}C_{10}$

11) $4 \cdot {}_{18}C_{11}$

12) ${}_{20}C_{16} + 1$

13) $\frac{{}_{20}C_5}{8}$

14) $-6 + {}_{19}C_5$

Critical thinking questions:15) Explain why ${}_nC_2 = {}_nC_{n-2}$

16) Write a combination that equals 12345